

Soumyadeb Tripathy

LinkedIn: [linkedin.com/in/soumyadeb-tripathy](https://www.linkedin.com/in/soumyadeb-tripathy)

GitHub: github.com/Rexy-5097

Portfolio: proof-of-work-roan.vercel.app

Email: soumyadeb043@gmail.com

Mobile: +91 7047660688

LeetCode: leetcode.com/u/ApexRaptor_5097

SKILLS SUMMARY

Languages:	Python, Rust, TypeScript, C++, SQL, JavaScript
Frameworks & Libraries:	Next.js, React, Astro, Node.js, FastAPI, PyTorch, scikit-learn, pandas, NumPy, tree-sitter, petgraph
Databases:	PostgreSQL, PL/pgSQL, Redis, Supabase, Parquet
Cloud / DevOps:	Docker, GitHub Actions (CI/CD), Linux, Vercel, Render, Supabase
Developer Tools:	Git, GitHub, Cargo, pytest, Playwright, Criterion
Core Subjects:	Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, System Design

PROJECTS

Cartograph – Cross-Language Code Architecture Engine | [GitHub](#) Aug 2026 – Present

- Built a **Rust** static-analysis engine (**6 crates, 15K LOC**) that links **TypeScript, TSX and Python** into one cross-language call graph, tracing any UI action to the database tables it hits.
- Benchmarked on **Sentry’s 3.1M-line codebase**: parsed **16,844 files in 8.5 s**, extracting **178K symbols** and **637K call sites**, then resolved **186 ORM models to 142 tables across 8,601 edges**.
- Attached confidence and provenance to every edge, **declining ambiguous matches rather than emitting false ones**; enforced by **335 tests** and a clippy-denied, unsafe-forbidden CI gate.

Tech Stack: Rust, tree-sitter, petgraph, Criterion, Cargo Workspaces, GitHub Actions

AdityaNet – Verifiable Research Platform, Aditya-L1 Solar X-ray Archive | [GitHub](#) | [Live](#) Jul 2026 – Present

- Built a **Python** pipeline parsing ISRO **Aditya-L1 SoLEXS/HEL10S** X-ray products under a fail-loud contract into a **digest-addressed (SHA-256) dataset — 7 tables, 1,985 Parquet files, 569 MiB** over 28 months.
- Ran a **pre-registered benchmark** (fixed seed, day-block bootstrap CIs) on a **192,541-minute chronological hold-out**, since random splits leak at **0.80 autocorrelation**. Learned models (**ROC-AUC 0.966**) **do not beat** a single count-rate threshold (**0.954**).
- Made a wrong figure unpublishable: a **CI gate re-reads every artifact and fails the build on drift**, and a **PolicyLeakageError gate** structurally refuses test-derived policy after I proved and voided my own leaked thresholds; 18 Astro routes ship **0 KB JavaScript**.

Tech Stack: Python, pandas, scikit-learn, Parquet, TypeScript, Astro, React, GitHub Actions, Render

FurnitureOps – Concurrency-Safe Inventory Backend | [GitHub](#) | [Live](#) Feb 2026 – Jul 2026

- Cut oversells from **500 to 0** across **1,000 purchase attempts from 50 concurrent workers** against 500 units, by routing every decrement through a **PL/pgSQL procedure** row-locking with **SELECT ... FOR UPDATE**.
- Sustained **~7,000 ops/s at 2.4 ms median** while writing an **append-only audit row in the same transaction** — correctness enforced in the database, not in application code.
- Decoupled purchases behind a **Redis queue** with bounded retries, a circuit breaker and a dead-letter queue; **idempotency keys** drop duplicate requests and **Postgres Row-Level Security** gates admin writes.

Tech Stack: TypeScript, Next.js 14, PostgreSQL, PL/pgSQL, Redis (Upstash), Supabase, Playwright, Vercel

ACHIEVEMENTS

Solved **543 problems on LeetCode** — **285 Medium** and **117 Hard** — publishing **149 curated C++ solutions** with complexity analysis. Apr 2026 – Present

EXTRACURRICULAR ACTIVITIES

Built a containerized **OpenEnv reinforcement-learning environment** for agent-driven PR review — FastAPI action/reward loop with ground-truth bug labels; submitted to Meta’s hackathon. Apr 2026

EDUCATION

Lovely Professional University

Bachelor of Technology in Computer Science and Engineering — CGPA: 7.19

Punjab, India

Aug 2024 – Jun 2028

Kishorenagar Sachindra Siksha Sadan

Senior Secondary (Class XII) — Science

Odisha, India

2023

Gap year (2023–24): self-directed study of data structures & algorithms, system design and backend development.